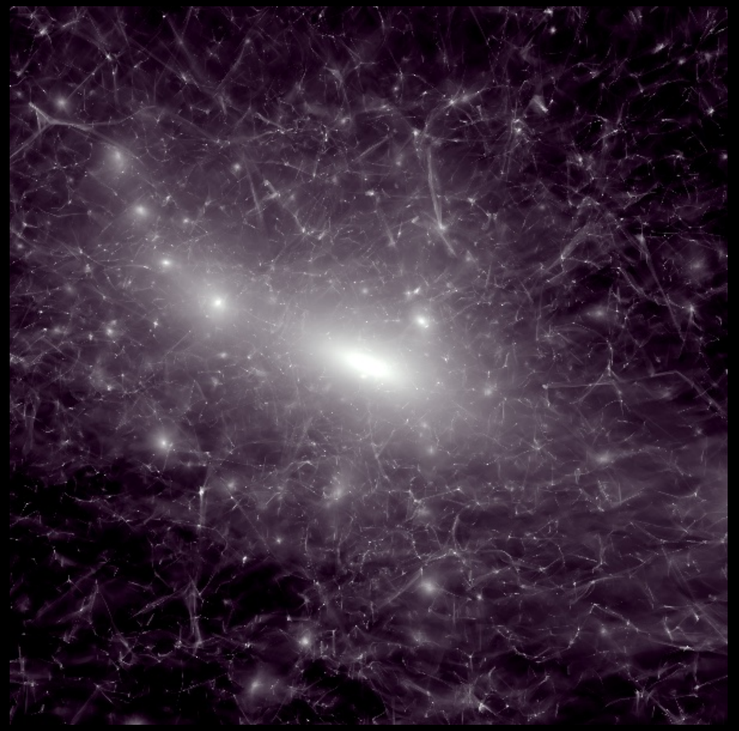
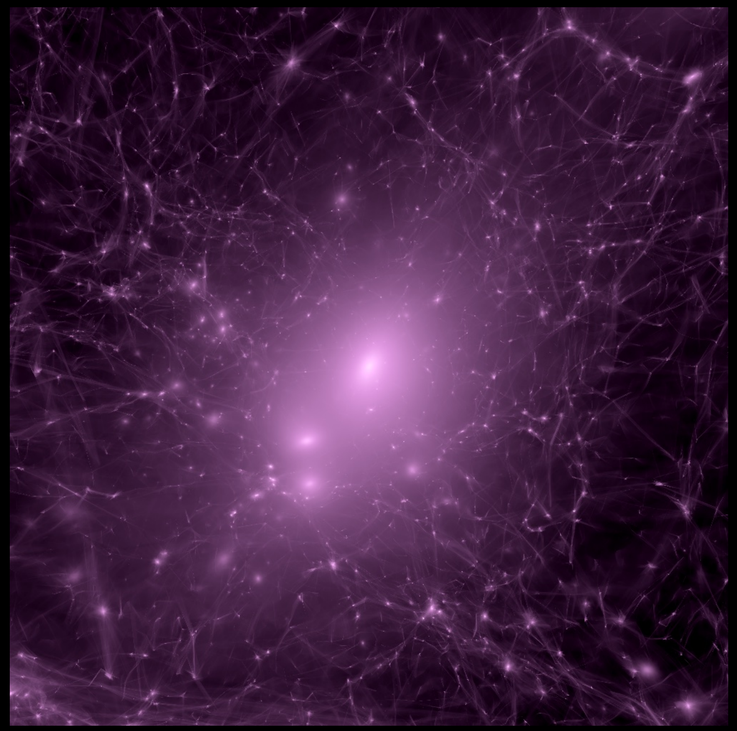
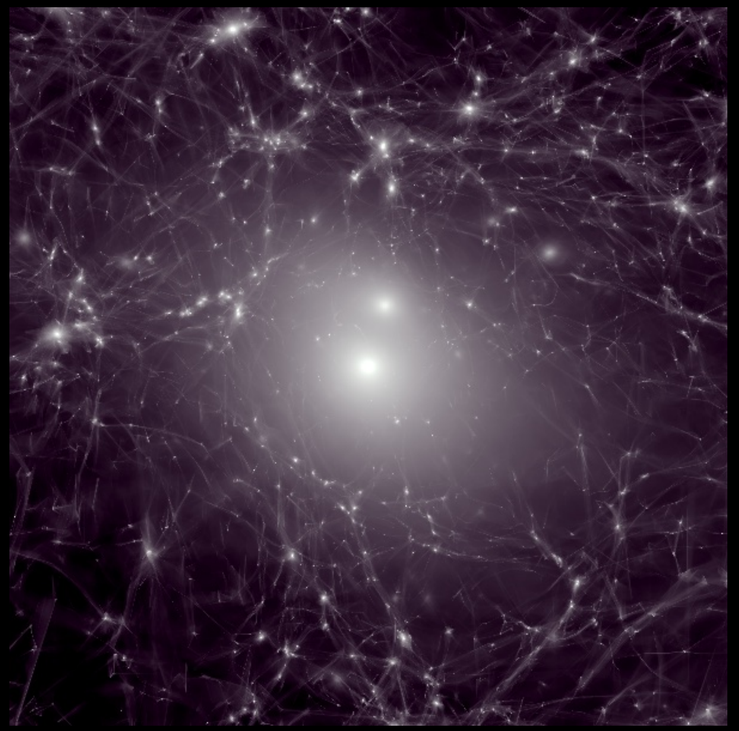
AstroReach SD: Dark Matter & Python







THE PREUSS SCHOOL

Week 4: Dark Matter Simulations
1/24/2025

UC San Diego

Dark Matter Simulations

If we can’t see dark matter, how do we make predictions about it?

- **Simulation**: “imitation of a situation or process”
- Dark matter simulations “imitate” the formation of structure in our universe
- These simulations are run on supercomputers for millions of CPU hours
- Simulations produce beautiful results, which we will visualize!

N-Body Simulations

The simplest approach:

- **N-body simulations** evolve a collection of N particles under gravity
- Since dark matter gravitates, this works reasonably well for large N
- There is a tradeoff between resolution and computational expense
- The first N-body simulations used N ~ 100; today, we can use N ~ 1 trillion!

N-Body Simulations

The first direct N-body simulation

THE ASTROPHYSICAL JOURNAL

AN INTERNATIONAL REVIEW OF SPECTROSCOPY AND ASTRONOMICAL PHYSICS

VOLUME 94 NOVEMBER 1941 NUMBER 3

ON THE CLUSTERING TENDENCIES AMONG THE NEBULAE

II. A STUDY OF ENCOUNTERS BETWEEN LABORATORY MODELS OF STELLAR SYSTEMS BY A NEW INTEGRATION PROCEDURE

ERIK HOLMBERG

Replace gravity by light!

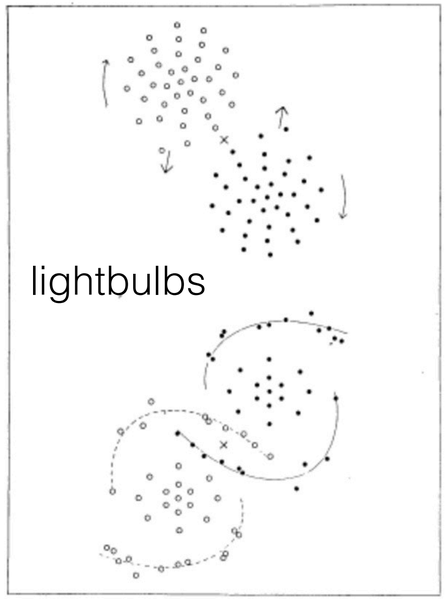


FIG. 4a

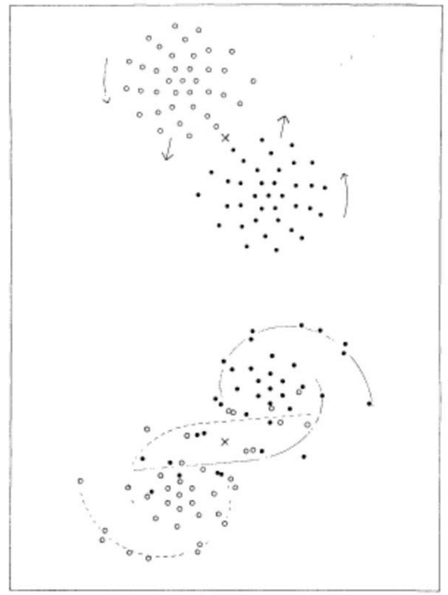


FIG. 4b

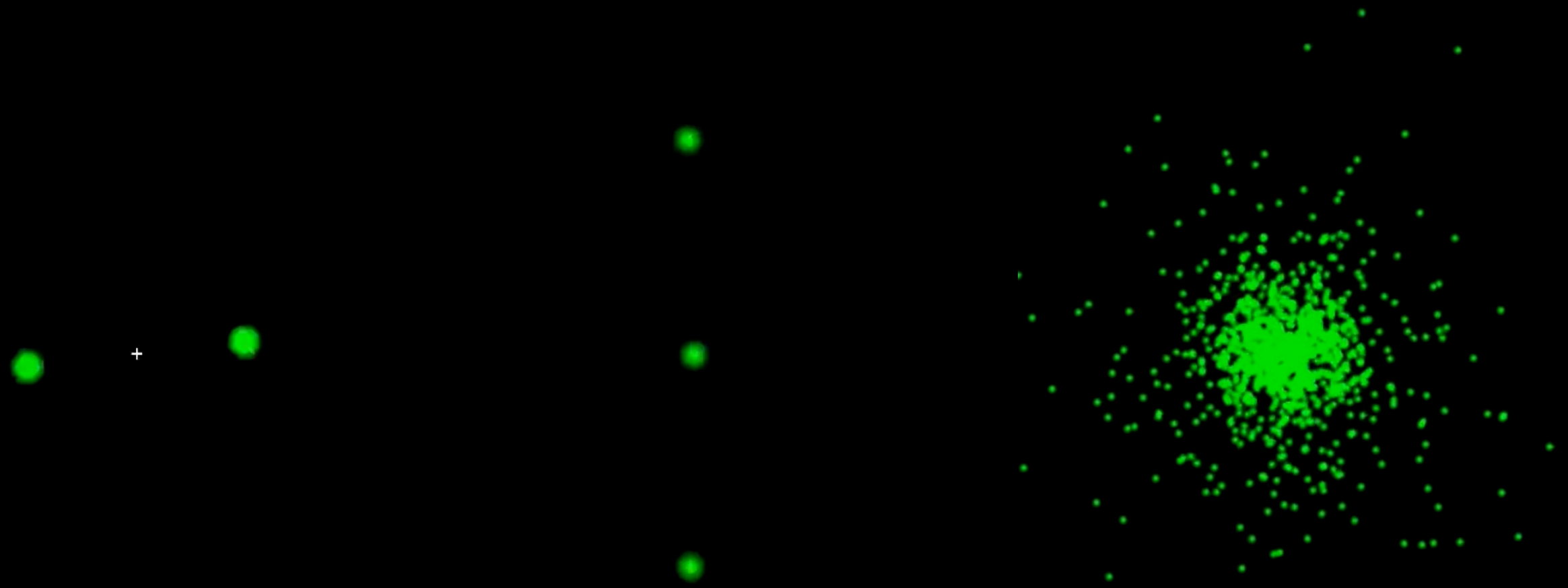
FIG. 4a.—Tidal deformations corresponding to parabolic motions, clockwise rotations, and a distance of closest approach equal to the diameters of the nebulae. The spiral arms point in the direction of the rotation.

FIG. 4b.—Same as above, with the exception of counterclockwise rotations. The spiral arms point in the direction opposite to the rotation.

2-body problem

3-body problem

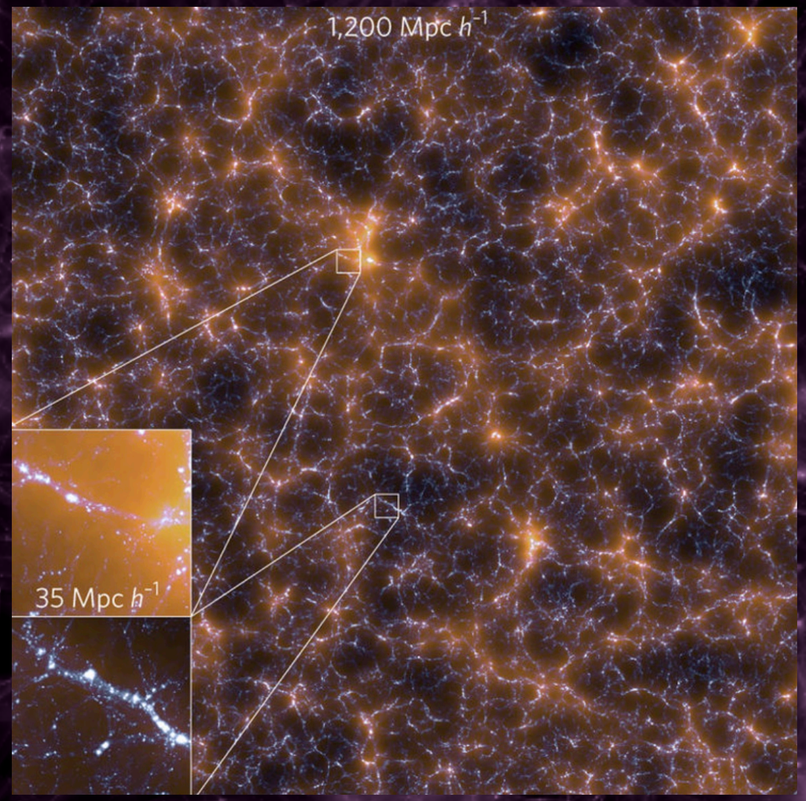
N-body problem



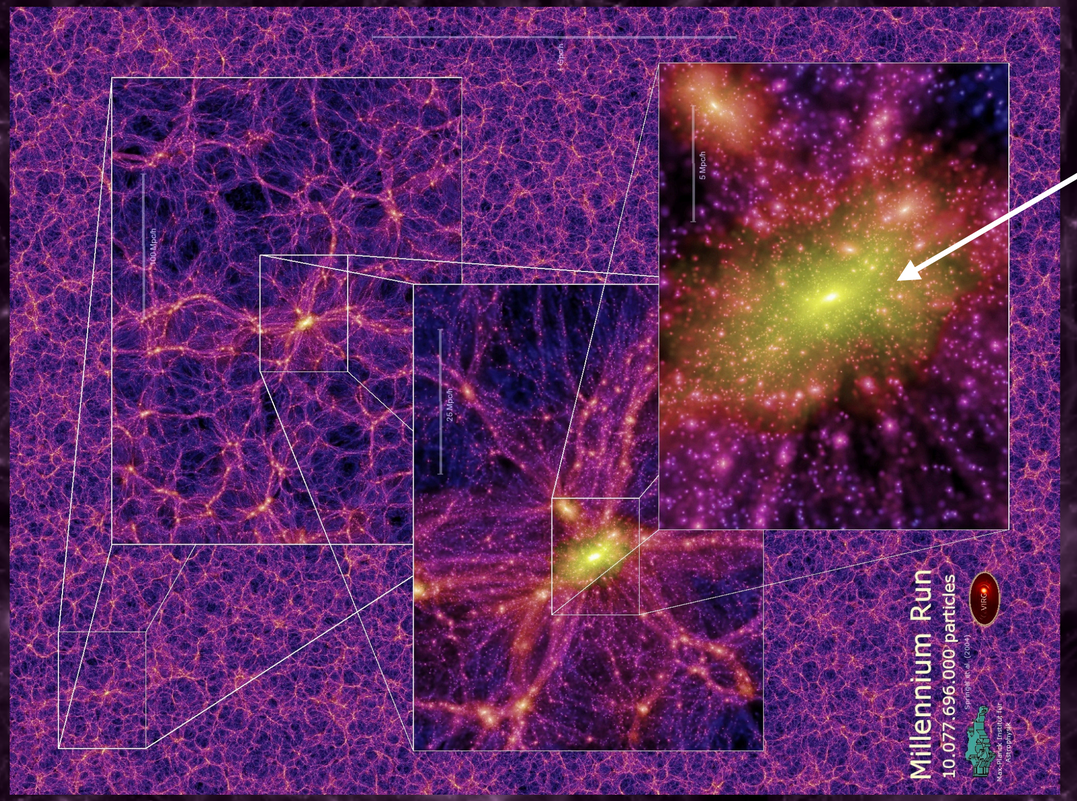
$$\vec{F}_i = \sum_j -G\frac{M_i M_j}{r_{ij}^2}\hat{r}_{ij}$$

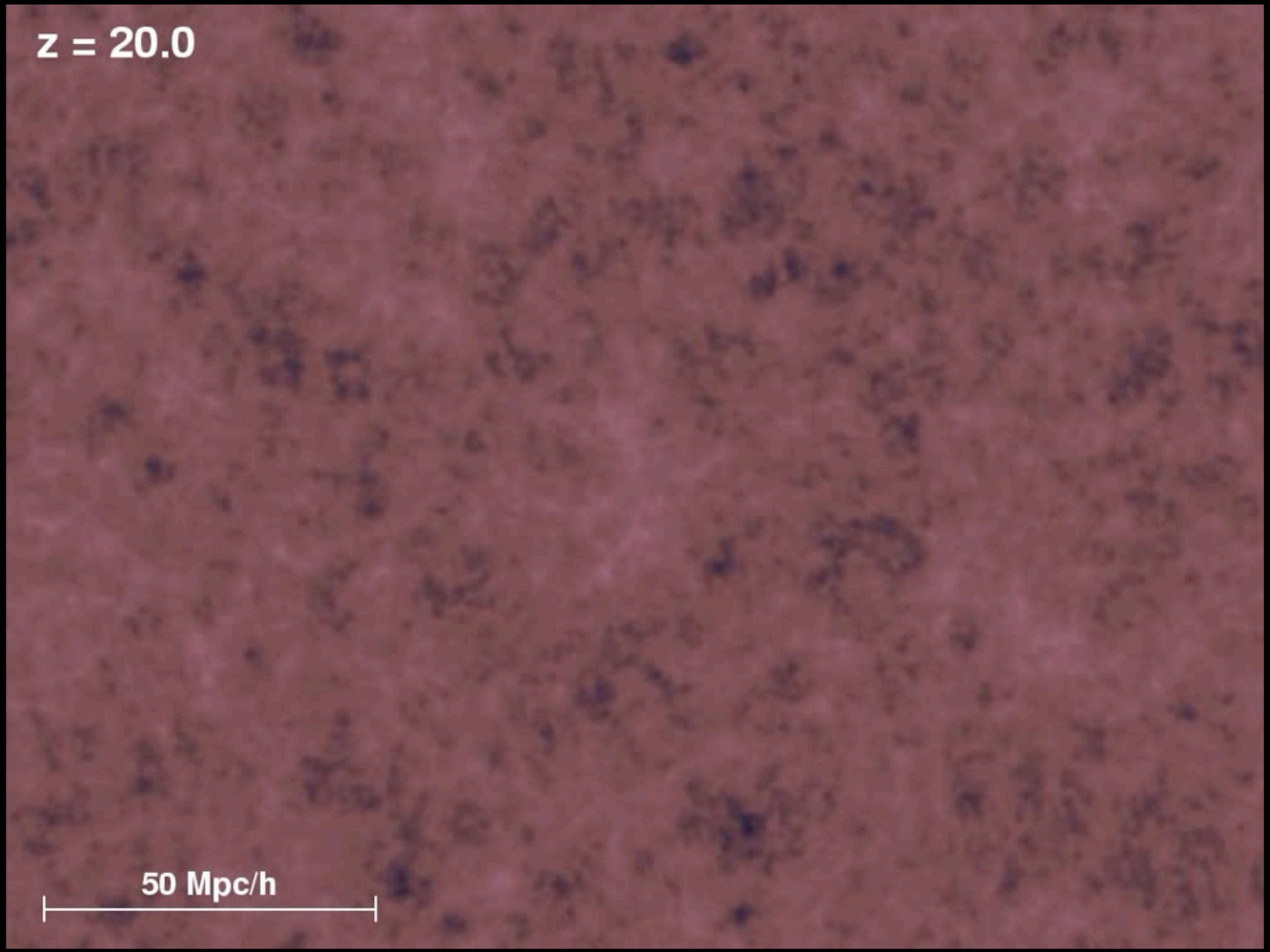
N-Body Simulations

Large volume



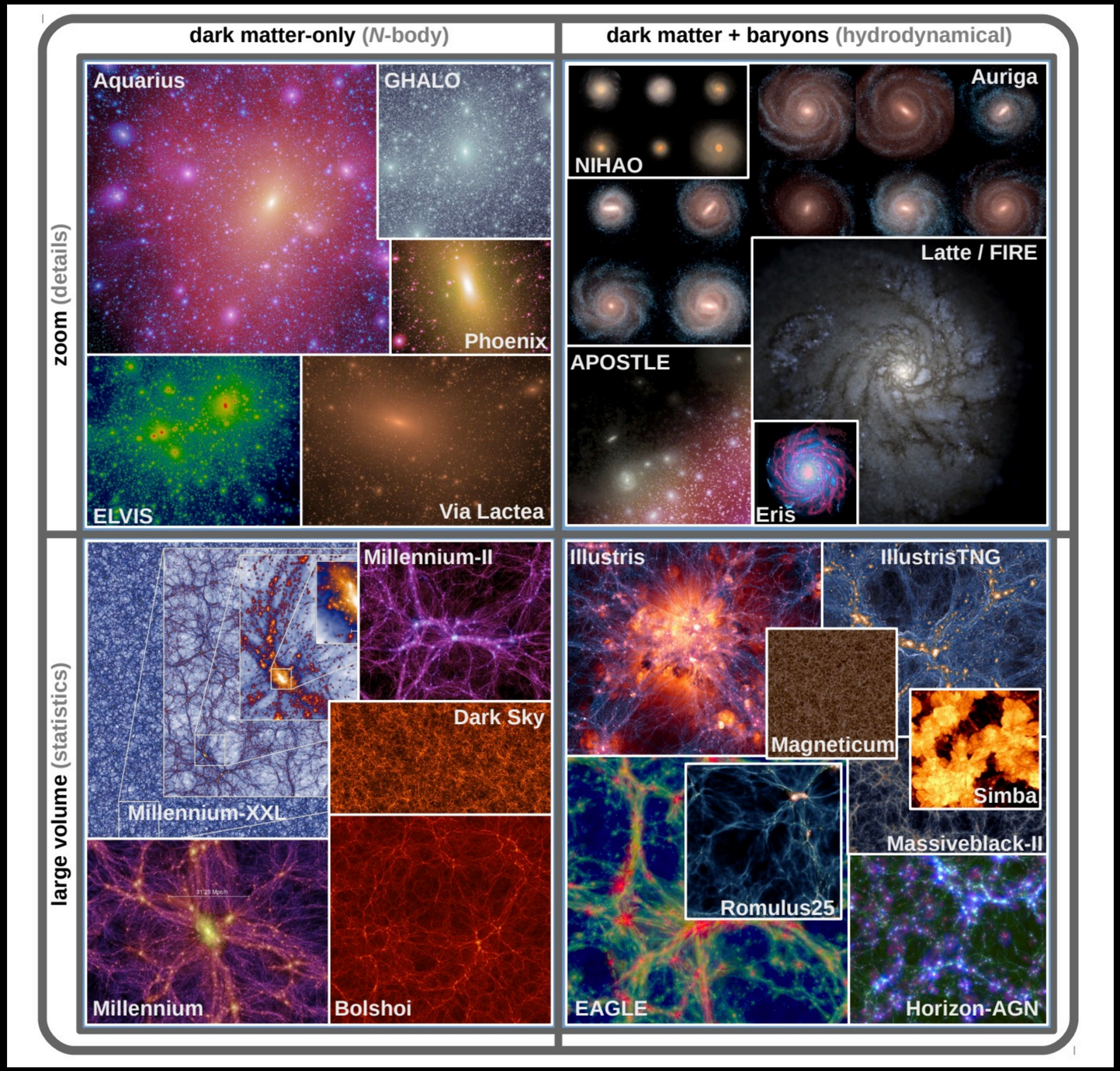
Smaller volumes




z = 20.0
50 Mpc/h

dark matter only
small volume

dark matter only
large volume



all matter
small volume

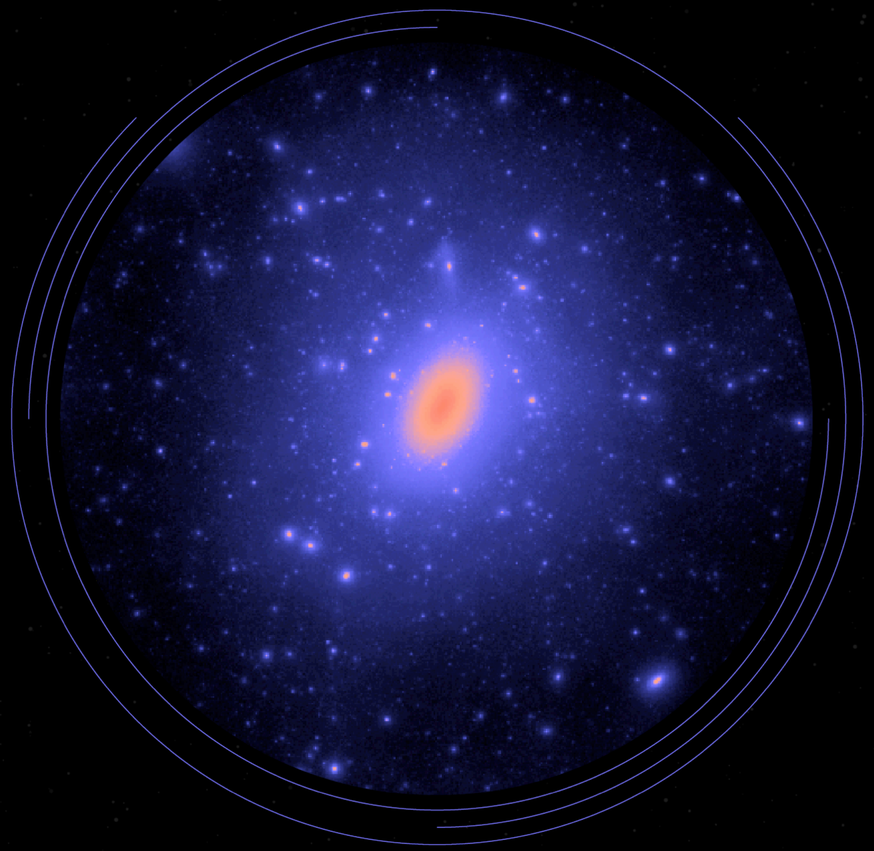
all matter
large volume

Using Simulations to Study Dark Matter

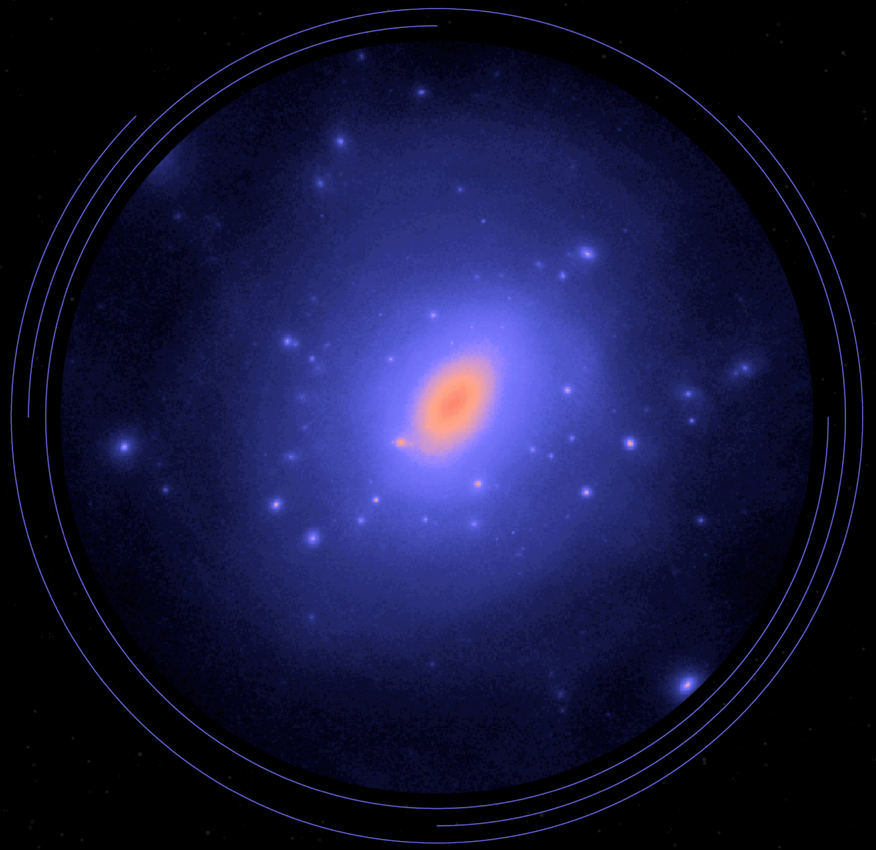
The outcome of a simulation depends on what we assume about dark matter:

- “Cold dark matter” gravitates, moves slowly, and clumps on all scales
- “Warm dark matter” gravitates, moves quickly, and clumps on large scales
- “Self-interacting dark matter” gravitates and collides with itself
- “Fuzzy dark matter” is so light that it’s wave-like instead of particle-like

Cold Dark Matter

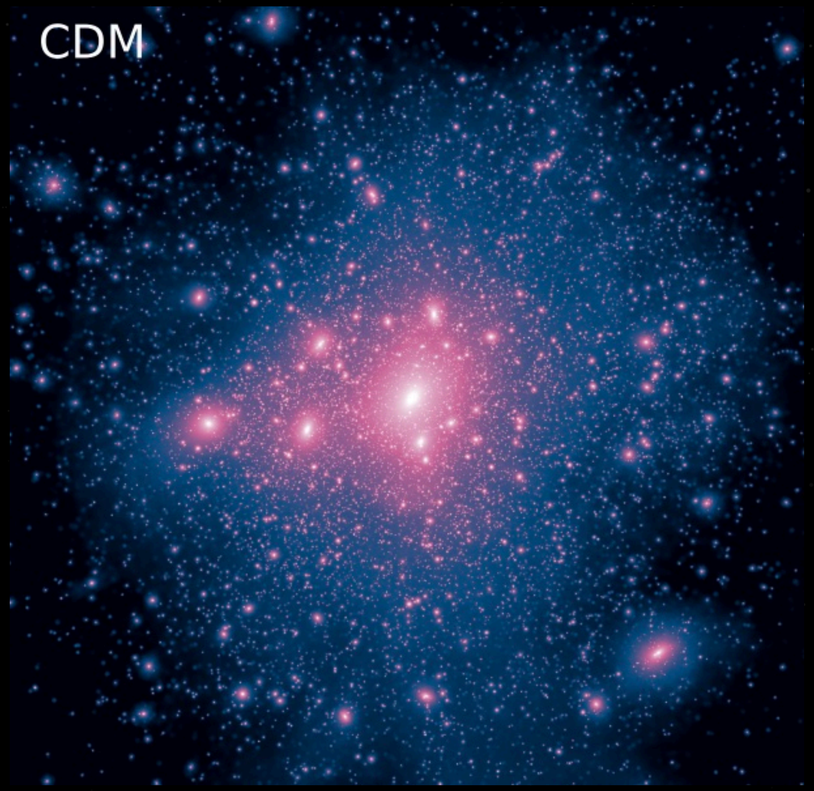


Warm Dark Matter

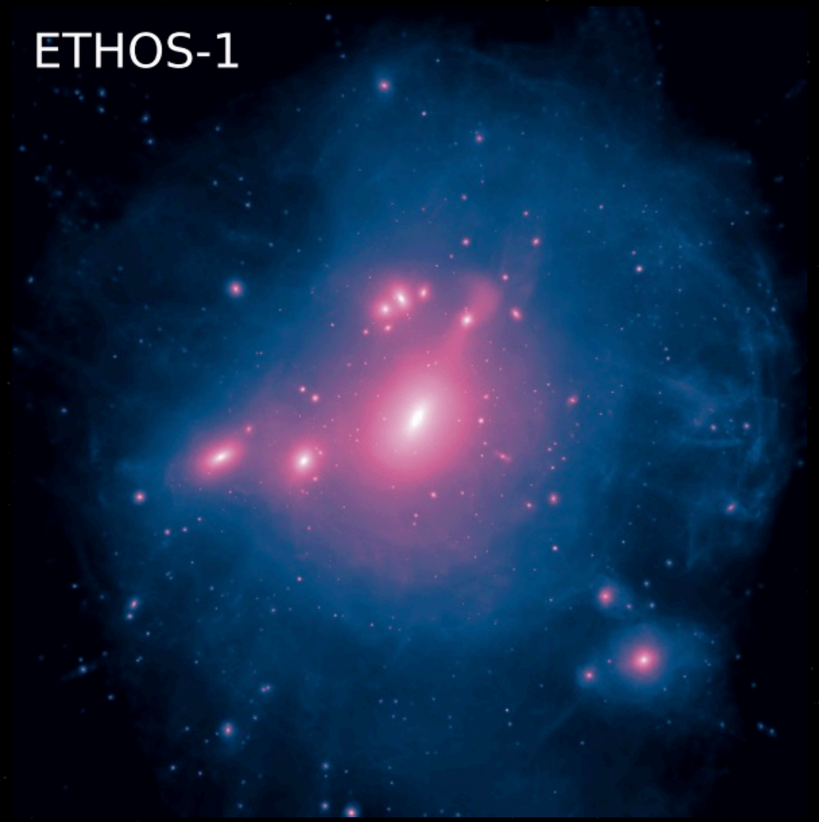


Source: Bullock & Boylan-Kolchin 2017 / Simulations by V. Robles, T. Kelley, and B. Bozek+

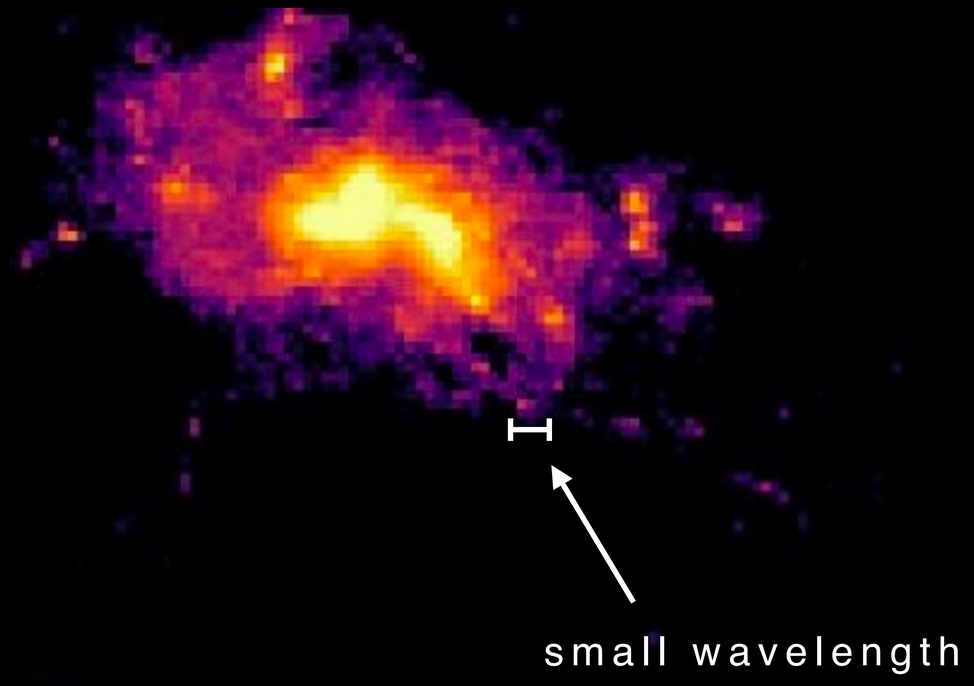
Cold Dark Matter



Interacting Dark Matter

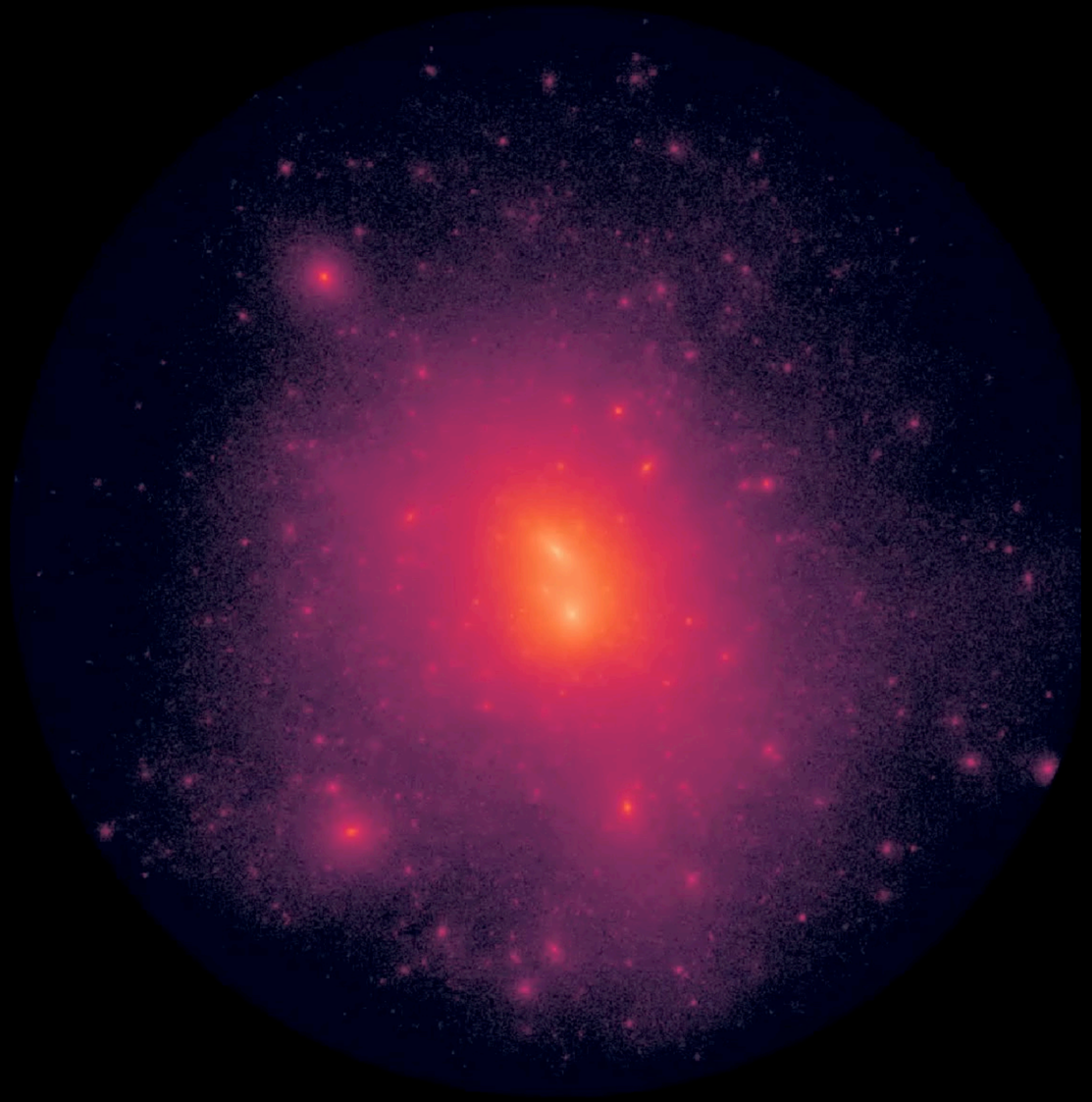


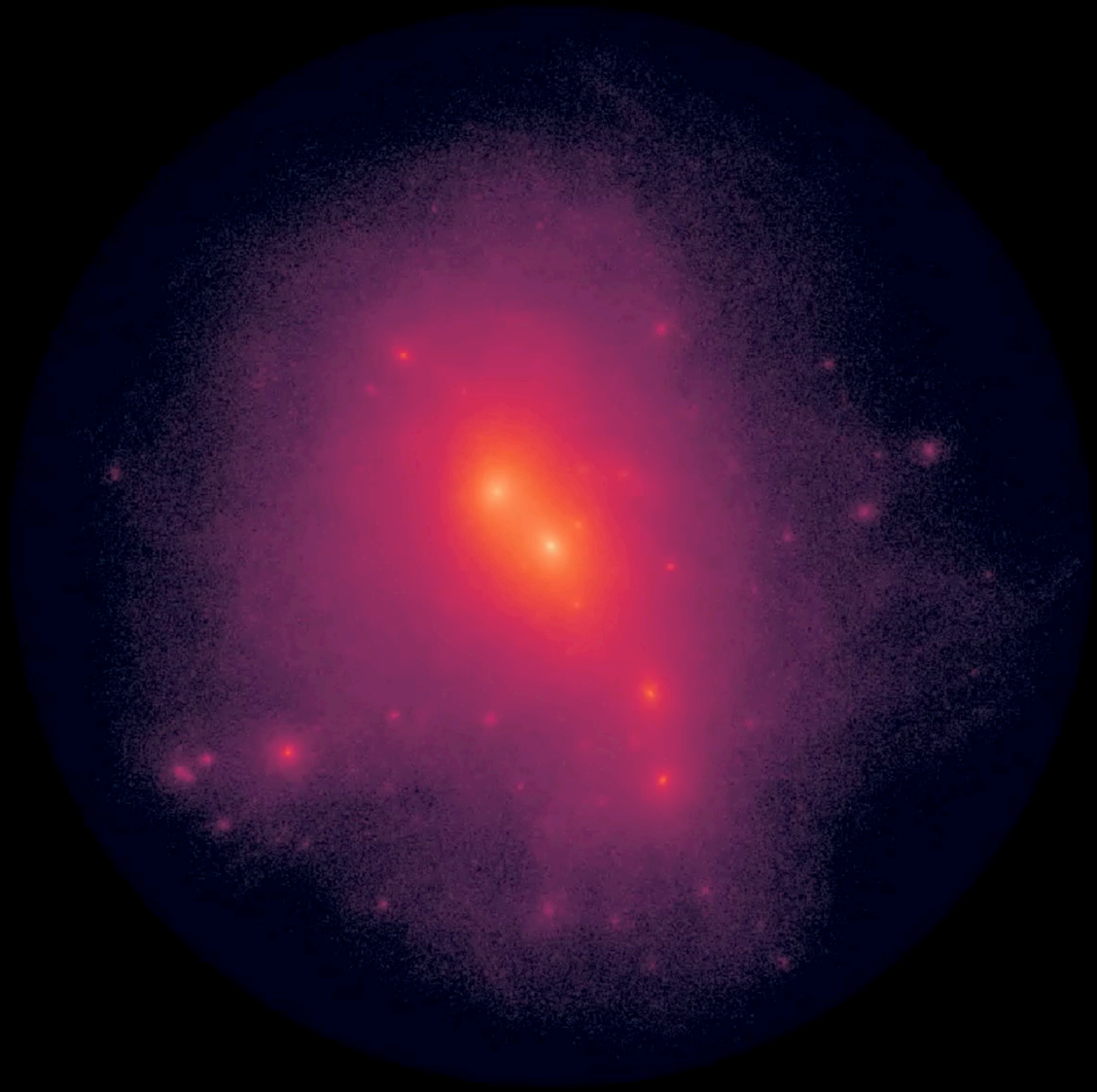
Cold Dark Matter



Fuzzy Dark Matter







SDSC SAN DIEGO SUPERCOMPUTER CENTER
at the UNIVERSITY OF CALIFORNIA SAN DIEGO
COMPUTING WITHOUT BOUNDARIES



- *Compute*: >2,500 CPUs (>100k cores) + 52 GPUs for accelerated computing
- Storage: >50 PB of storage with high-performance parallel file systems
- *Connectivity*: 100 GB/second link to internet and other research networks

Discussion

Discuss the following topics with your neighbor:

- What do you find most interesting/surprising/confusing from today's material?
- What questions do you have about the universe based on today's material?